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Large Magneto-Current Effect in the Electrochemical Detection of Oxalate in Aqueous Solution

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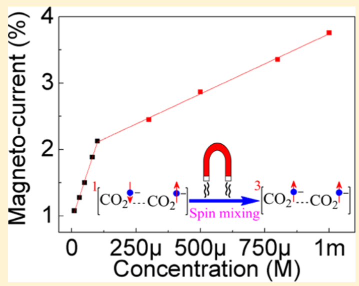
Supporting Information

ABSTRACT: Herein, we first report an interesting observation of a large magneto-current (MC) of nearly 30% based on the electrochemical oxidation of oxalate in aqueous solution at room temperature. The large MC is ascribed to spin-dependent oxidation of oxalate. Both singlet and triplet radical pairs are generated during the electrochemical oxidation of oxalate. An applied magnetic field could accelerate the spin evolution of a singlet radical pair into its triplet state. The triplet radical pair has a much larger dissociation probability than the singlet radical pair because of the Pauli exclusion principle. Thus, enhancing the triplet yield could largely increase the oxidation rate of oxalate and generate significant MC. The significant MC effect provides a new approach to study spin evolution of radical pairs in electrochemical cells. Furthermore, an effective analytical method for oxalate determination can be developed based on the significant MC effect.



1. INTRODUCTION

Magnetic fields can alter the rate, yield, or product distribution of chemical reactions, which is known as spin chemistry.[1−5] At present, the study of magnetic field effects in electrochemical reactions is one of the most interesting problems of spin chemistry.[6−9] This develops a new paradigm: spin-dependent electrochemistry. The experimental implementation of spin-dependent electrochemistry was first described in a 2013 paper.[10] After that a number of different applications of spin-dependent electrochemistry concept in a variety of different systems emerged.[6−9,11] Examples include chiral-induced spin-polarized photoelectrons to enhance the photocurrent in photo-electrochemical systems[9,12] and electron spin polarization achieved through chiral molecules leading to a more efficient water-splitting process.[13] Different from chiral-induced spin-dependent electrochemistry reported currently, here we present a study of magnetic field effects on electrical current, namely, magneto-current (MC) based on the electrochemical oxidation of oxalate. The significant MC is ascribed to the spin-dependent oxidation of oxalate, which is totally different from the chiral-induced spin selectivity effect in electrochemical systems reported previously.[6−9,11] The spin-dependent oxidation of oxalate is essentially attributed to magnetic field-induced spin mixing between singlet and triplet radical pairs. During the oxidation process of oxalate, both singlet and triplet radical pairs are generated. An applied magnetic field could facilitate the spin evolution from singlet to triplet in spin-correlated radical pairs[1,14−17] and largely raise the triplet yield. Because of the triplet radical pair owning larger dissociation probability than the singlet radical pair, the increase of triplet yield could promote the oxidation process of oxalate and enhance the oxidation current. Therefore, the application of a magnetic field can increase the oxidation current of oxalate by enhancing the triplet yield through magnetic field-facilitated intersystem crossing process[1,3,15−18] and generate positive MC. Consequently, we investigate the possibility to develop a new analytical method for oxalate determination via the significant MC effect.

High oxalate concentration in the blood or urine accompanies a number of diseases including renal failure, intestinal diseases, and hyperoxaluria.[19] It may also lead to the development and formation of renal and urinary stones.[20] Thus, sensitive and convenient detecting methods are necessary for the determination of oxalate content in food chemistry and in clinical analysis. Various methods such as fluorometry,[21,22] colorimetry,[23,24] gas chromatography,[25] ion chromatography,[26,27] electrogenerated chemiluminesce,[19] and enzymatic methods[28,29] have been used to quantify oxalate. However, most of these methods often require complicated sample pretreatment (e.g., a preliminary separation of oxalate from the biological matrix)[28,29] and expensive equipment such as gas chromatographs.[22,25−27] Therefore, a better method for direct determination of oxalate in biological samples is needed.

2. RESULTS AND DISCUSSION

Figure 1 compares the MC based on the oxidation of 0.1 M oxalate solution at different applied electrode potentials. We found that applying a magnetic field could dramatically increase the oxidation current, leading to positive MC. The peak values of MC (MC_{max}) were evaluated to be about 2.9,

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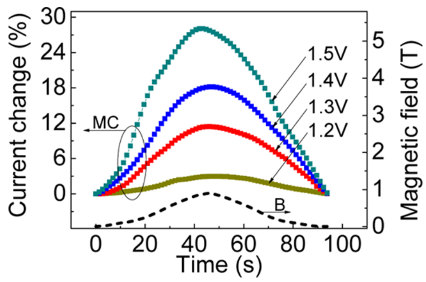
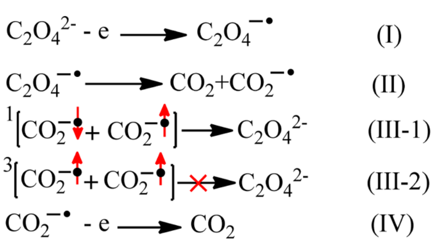


Figure 1. Current change is shown as a function of time in a triangular wave of magnetic field at different applied electrode potentials for a solution containing 0.1 M $Na_2C_2O_4$ and 0.1 M NaH_2PO_4.

11.4, 18.2, and 28.1% at an applied electrode potential of 1.2, 1.3, 1.4, and 1.5 V (vs Ag/AgCl), respectively, in a magnetic field of about 0.9 T (Figure 1). Generally, the MC in solution can be generated by spin-dependent chemical or electrochemical reactions[3,15,16,30] or magneto-convection effects induced by the Lorentz force effect.[31] In this system, the MC measurements were performed at the zero angle condition between magnetic field and charge transport direction (shown in Supporting Information Figure S1), which minimizes the magneto-convection effect induced by Lorentz force. In addition, when the applied electrode potential is set less than 1.1 V for 0.1 M oxalate solution, the MC is too low to be observed (Figure S2b), presumably because of the slow oxidation rate of oxalate (Figure S2a). Furthermore, the amplitude of MC increases as the applied electrode potentials increase. Because increasing the applied electrode potentials can increase the oxidation rate of oxalate (Figure S2a), thus the amplitude of MC increases. These results further suggest that the positive MC is generated by magnetic field-sensitive oxidation process of oxalate and exclude the MC effects.

In order to reveal the positive MC in this system, we propose the relevant electrochemical oxidation mechanism of oxalate[32−37] shown in Scheme 1.

Scheme 1. Proposed Mechanism Is Shown for Electrochemical Oxidation Process of Oxalate



According to Scheme 1, the oxidation of oxalate is characterized with the electron transfer (reactions I and IV), the dissociation of $C_2O_4^{-\bullet}$ into CO_2 and $CO_2^{-\bullet}$ radical anions (reaction II) and the recombination of two $CO_2^{-\bullet}$ radicals into $C_2O_4^{2-}$ anions (reaction III-1). The reaction steps during the oxidation process of oxalate are consecutive. Thus, the increase of reaction rate in any reaction step causes an enhancement in the overall reaction rate.

Before applying an external magnetic field, the oxidation process of oxalate stays in a dynamic balance with a stable electrical current over time at a constant applied electrode potential (Figure S2b−f). Essentially, the necessary C−C′ bond cleavage in $C_2O_4^{-\bullet}$ radical[32−34] results in CO_2 and $CO_2^{-\bullet}$ radical anions (Scheme 1) adjacent to the working electrode surface. In addition, the intermediate radicals produced by electrochemical oxidation of oxalate were characterized with the absorption spectroscopy, as shown in Figure 2b. The peak at 235 nm corresponds to the characteristic absorption spectrum of carboxyl radical anion $CO_2^{-\bullet}$.[38−40] This confirms the generation of carboxyl radical anion $CO_2^{-\bullet}$ during the oxidation process of oxalate. In the reaction zone, the freely diffusing $CO_2^{-\bullet}$ radical anions could encounter one another and the electron spins located on individual radicals will instantly interact with each other and form radical pairs $[CO_2^{-\bullet}\cdots CO_2^{-\bullet}]$ (Figure 2a). The formed radical pairs have two spin configurations: singlet $^1[CO_2^{-\bullet}\uparrow\cdots CO_2^{-\bullet}\downarrow]$ and triplet $^3[CO_2^{-\bullet}\uparrow\cdots CO_2^{-\bullet}\downarrow]$. Statistically, the ratio of the generated singlet and triplet radical pairs is 1:3.[18] The two radicals in the singlet radical pairs $[CO_2^{-\bullet}\uparrow\cdots CO_2^{-\bullet}\downarrow]$ could easily react and recombine to generate oxalate ions ($C_2O_4^{2-}$). This delays the overall oxidation process of oxalate, temporarily reducing the oxidation current. However, the recombination of triplet radical pairs $^3[CO_2^{-\bullet}\uparrow\cdots CO_2^{-\bullet}\uparrow]$ is spin forbidden because of the Pauli exclusion principle.[18,41] Thus, most of the triplet radical pairs tend to dissociate into $CO_2^{-\bullet}$ radical anions. The $CO_2^{-\bullet}$ radical anion then goes through further electron transfer and engenders CO_2. The recombination and dissociation of radical pairs $[CO_2^{-\bullet}\cdots CO_2^{-\bullet}]$ during the oxidation process of oxalate can reach a dynamic balance with a stable electrical current under a constant applied electrode potential. An external magnetic field could disturb the dynamic balance of oxalate oxidation process by altering the singlet → triplet intersystem crossing rate[1,3,18] in spin-correlated radical pairs $[CO_2^{-\bullet}\cdots CO_2^{-\bullet}]$.

In general, the intersystem crossing process between singlet and triplet radical pairs can be driven by magnetic interactions in the radical pairs and by any applied magnetic field.[1,42,43] In our system, the radical pairs formed by random encounter of freely diffusing radicals are statistically 25% singlet and 75% triplet. As singlet and triplet born radical pairs show equal and opposite field responses, it is common to simply treat such radical pairs as being triplet born (effectively 50% of them shows a triplet response and 50% shows no response). The triplet radical pair has three sublevels (T_+, T_-, and T_0). Before applying an external magnetic field, all three triplet states can be converted to singlet which can then recombine to oxalate ions. The application of an external magnetic field causes Zeeman splitting of the triplet sublevels (T_+ and T_-).[15,44−46] Varying the field strength can enhance the energy gap (ΔE) between singlet (S) and triplet sublevels (T_+ and T_-) and largely reduces the $T_\pm$ → S interconversion rate. This is equivalent to the increase of S → $T_\pm$ interconversion rate induced by an applied magnetic field. As a consequence, the yield of triplet radical pairs increases with the application of a magnetic field. Furthermore, the singlet radical pairs $^1[CO_2^{-\bullet}\uparrow\cdots CO_2^{-\bullet}\downarrow]$ could recombine and generate oxalate ions ($C_2O_4^{2-}$). This results in a delay of the overall reaction rate and temporarily reduces the electrical current. However, the recombination of triplet radical pairs $^3[CO_2^{-\bullet}\uparrow\cdots CO_2^{-\bullet}\uparrow]$ is spin forbidden because of the Pauli exclusion principle. Thus, the triplet pairs have a much larger dissociation probability than the singlet pairs. As a result, the enhancement of the triplet yield could accelerate the overall oxidation process of oxalate and cause an increase on the oxidation current. Consequently, the application of an external magnetic field can

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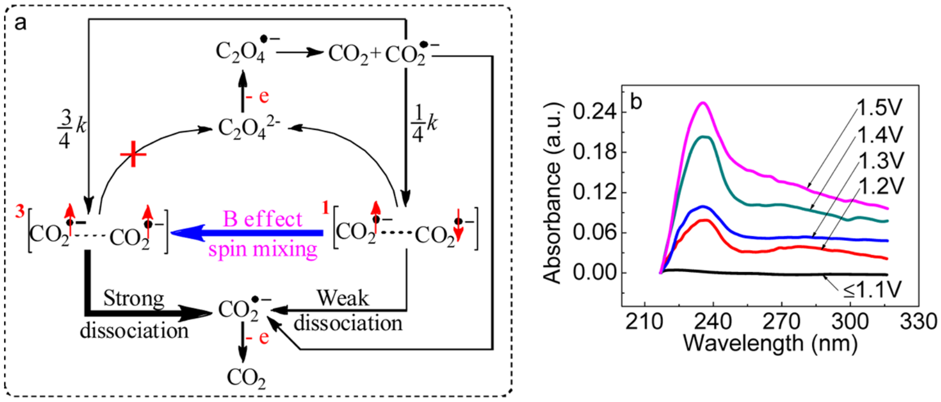

Figure 2. (a) Schematic diagram to show spin-dependent oxidation of oxalate-induced positive MC in aqueous solutions. (b) Absorption spectra of intermediates produced by electrochemical oxidation of 0.05 M sodium oxalate solution at different applied electrode potentials.

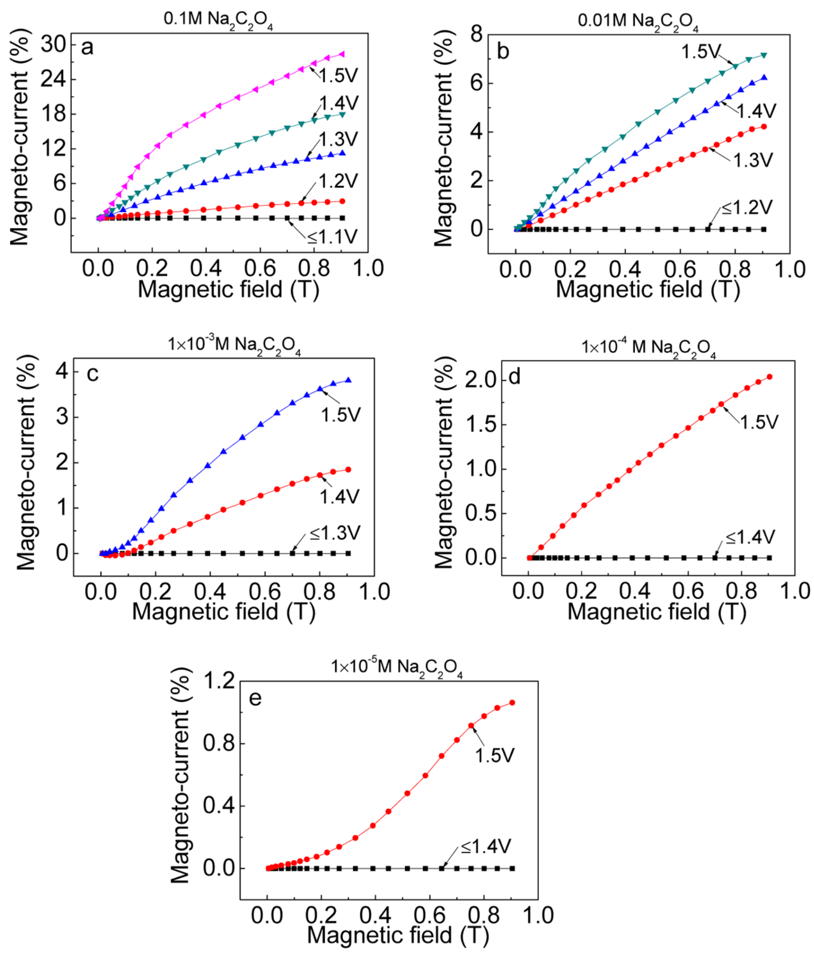

Figure 3. MC is shown at different applied electrode potentials for different concentrations of sodium oxalate dissolved in water. (a) 0.1 M $Na_2C_2O_4$; (b) 0.01 M $Na_2C_2O_4$; (c) 1×10^{-3} M $Na_2C_2O_4$; (d) 1×10^{-4} M $Na_2C_2O_4$; and (e) 1×10^{-5} M $Na_2C_2O_4$.

increase the oxidation current by facilitating the singlet → triplet interconversion and positive MC is generated.

To further confirm the magnetic field-accelerated intersystem crossing process in radical pairs $[CO_2^{-\bullet}\cdots CO_2^{-\bullet}]$ is the mechanism responsible for the positive MC based on the oxalate system, we studied the MC in different concentrations of radical pairs by altering the oxalate concentration and applied electrode potentials. Figure 3 shows the MC at different applied electrode potentials for oxalate solution in the range of 0.1 to 1×10^{-5} M.

First, when the applied electrode potentials are less than a certain value (e.g., 1.1 V in 0.1 M oxalate, 1.2 V in 0.01 M oxalate, 1.3 V in 1×10^{-3} M oxalate, 1.4 V in 1×10^{-4} M, and 1×10^{-5} M oxalate), the oxidation rate of oxalate is very low (Figure S3), few intermediate products ($CO_2^{-\bullet}$) are generated. This results in very few radical pairs $[CO_2^{-\bullet}\cdots CO_2^{-\bullet}]$ in the reaction zone. Thus, hardly any MC is observed (Figure 3a−e, black line) at low applied potentials or in low oxalate concentration. These results further rule out the Lorentz force and magneto-convection effects to be the main reason for

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the generation of MC. Besides, as applied electrode potentials increase, the MC amplitude increases accordingly in different concentrations of oxalate (Figure 3). Because increasing the applied electrode potentials and oxalate concentration could enhance the oxidation rate of oxalate, thus the concentration of intermediate products ($CO_2^{-\bullet}$) in the reaction zone greatly increases. This is further confirmed by the increase in the absorption intensity of $CO_2^{-\bullet}$ radicals as applied electrode potentials increase (Figure 2b). The increase of the concentration of $CO_2^{-\bullet}$ could result in a larger encounter probability of $CO_2^{-\bullet}$ with each other. This leads to the formation of a higher concentration of radical pairs [$CO_2^{-\bullet}\cdots CO_2^{-\bullet}$]. As a consequence, the amount of radical pairs [$CO_2^{-\bullet}\cdots CO_2^{-\bullet}$] participate in the magnetic field-facilitated intersystem crossing to generate the triplet radical pairs is enhanced. Thus, the amplitude of MC increases as the concentration of oxalate increases, as shown in Figure 4. The

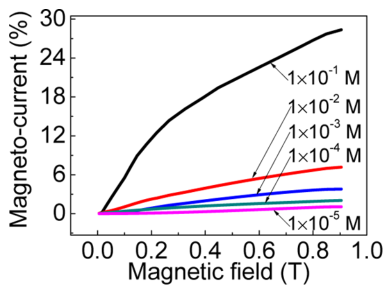

Figure 4. MC is shown for different concentrations of sodium oxalate dissolved in water at 1.5 V applied electrode potential.

MC based on different concentrations of oxalate with different applied electrode potentials further confirms that the magnetic field-accelerated intersystem crossing process in radical pairs [$CO_2^{-\bullet}\cdots CO_2^{-\bullet}$] is the main mechanism responsible for the positive MC observed in the present system.

In the end, we investigated the possibility to develop a new analytical method for oxalate detection based on the significant MC effect. Figure 5 shows that the maximal amplitude of MC

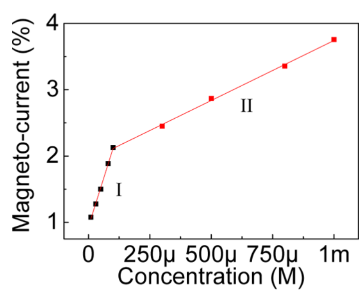

Figure 5. Amplitude of MC is linearly related to the concentration of oxalate over the region 1×10^{-5} to 1×10^{-4} M and 1×10^{-4} to 1×10^{-3} M in the magnetic field of 0.9 T.

(MC_{max}) is linearly related to the concentration of oxalate over two regions: region I, from 1×10^{-5} to 1×10^{-4} M (correlation coefficient = 0.99737, slope = 1.18×10^4, $\delta_s = 0.03 \times 10^4$) and region II, from 1×10^{-4} to 1×10^{-3} M (correlation coefficient = 0.99801, slope = 1.81×10^3, $\delta_s = 0.04 \times 10^3$). The linear correlation slope between MC amplitude and oxalate concentration in region I is about 6.5 times larger than that in region II (Figure 5). In general, the amplitude of MC can be expressed as follows

$$MC = \frac{I_B - I_0}{I_0} \times 100\% = \frac{\Delta I_B}{I_0} \times 100\% \quad (1)$$

where ΔI_B is proportional to the concentration of radical pairs [$CO_2^{-\bullet}\cdots CO_2^{-\bullet}$] (Figure 1) and I_0 is proportional to the concentration of oxalate (Figure S3). It should be noted that when the concentration of oxalate decreases, the concentration of generated radical pairs [$CO_2^{-\bullet}\cdots CO_2^{-\bullet}$] will also decrease. Thus, both ΔI_B and I_0 decrease as the concentration of oxalate drops. Therefore, the amplitude of MC does not decrease as fast as the concentration of oxalate. Consequently, the linear correlation slope between MC amplitude and oxalate concentration in the low concentration region is larger than that in the high concentration region (the slope in region I is about 6.5 times larger than that in region II). Additionally, oxalate concentrations in normal blood range from about 1.7×10^{-5} to 3.9×10^{-5} M and in normal urine range from about 1.6×10^{-4} to 5.5×10^{-4} M.[19] Thus, the MC method has sufficient sensitivity for oxalate determination in these clinically interesting fluids. Furthermore, compared with current methods[21,23,25−28] for oxalate detection, this MC method offers simple sample pretreatment, low cost, and easy performance.

3. CONCLUSIONS

We report a large MC of nearly 30% in aqueous solution at room temperature based on the electrochemical oxidation of oxalate. The large MC is attributed to spin-dependent oxidation process of oxalate. During the electrochemical oxidation process of oxalate, singlet and triplet radical pairs are both generated. An applied magnetic field could facilitate the singlet → triplet interconversion through spin mixing. Because of the larger dissociation probability of the triplet pairs than the singlet pairs, the enhancement of triplet yield could result in an increase of the oxidation current. This leads to the generation of positive MC. By modulating the concentration of radical pairs through changing the concentration of oxalate and applied electrode potentials, we could tune the amplitude of MC based on the electrochemical oxidation of oxalate. The significant MC based on oxalate oxidation provides a new approach to study spin evolution of radical pairs in electrochemical cells. Furthermore, an effective analytical method for oxalate detection can be developed based on the significant MC effect.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.8b04193.

Experimental setup of MC measurement and the cyclic voltammograms of different concentrations of oxalate solution and MC with different applied electrode potentials are provided to exclude Lorentz force and magneto-convection effect in our system (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Steiner, U. E.; Ulrich, T. Magnetic field effects in chemical kinetics and related phenomena. *Chem. Rev.* **1989**, *89*, 51−147.
(2) Rodgers, C. T.; Hore, P. J. Chemical magnetoreception in birds: the radical pair mechanism. *Proc. Natl. Acad. Sci. U.S.A.* **2009**, *106*, 353−360.
(3) Gould, I. R.; Turro, N. J.; Zimmt, M. B. Magnetic field and magnetic isotope effects on the products of organic reactions. *Advances in Physical Organic Chemistry*; Academic Press, 1984; Vol. *20*, pp 1−53.
(4) Hore, P. J.; Mouritsen, H. The Radical-Pair Mechanism of Magnetoreception. *Annu. Rev. Biophys.* **2016**, *45*, 299−344.
(5) Beardmore, J. P.; Antill, L. M.; Woodward, J. R. Optical absorption and magnetic field effect based imaging of transient radicals. *Angew. Chem.* **2015**, *127*, 8614−8617.
(6) Mondal, P. C.; Fontanesi, C.; Waldeck, D. H.; Naaman, R. Field and chirality effects on electrochemical charge transfer rates: spin dependent electrochemistry. *ACS Nano* **2015**, *9*, 3377−3384.
(7) Fontanesi, C. Spin-dependent electrochemistry: A novel paradigm. *Curr. Opin. Electrochem.* **2018**, *7*, 36−41.
(8) Mondal, P. C.; Mtangi, W.; Fontanesi, C. Chiro-Spintronics: Spin-dependent electrochemistry and water splitting using chiral molecular films. *Small Methods* **2018**, *2*, 1700313.
(9) Mtangi, W.; Tassinari, F.; Vankayala, K.; Vargas Jentzsch, A.; Adelizzi, B.; Palmans, A. R. A.; Fontanesi, C.; Meijer, E. W.; Naaman, R. Control of Electrons' Spin Eliminates Hydrogen Peroxide Formation During Water Splitting. *J. Am. Chem. Soc.* **2017**, *139*, 2794−2798.
(10) Mishra, D.; Markus, T. Z.; Naaman, R.; Kettner, M.; Gohler, B.; Zacharias, H.; Friedman, N.; Sheves, M.; Fontanesi, C. Spin-dependent electron transmission through bacteriorhodopsin embedded in purple membrane. *Proc. Natl. Acad. Sci. U.S.A.* **2013**, *110*, 14872−14876.
(11) Kumar, A.; Capua, E.; Vankayala, K.; Fontanesi, C.; Naaman, R. Magnetless device for conducting three-dimensional spin-specific electrochemistry. *Angew. Chem.* **2017**, *129*, 14779−14782.
(12) Zhang, W.; Gao, W.; Zhang, X.; Li, Z.; Lu, G. Surface spintronics enhanced photo-catalytic hydrogen evolution: mechanisms, strategies, challenges and future. *Appl. Surf. Sci.* **2018**, *434*, 643−668.
(13) Mtangi, W.; Kiran, V.; Fontanesi, C.; Naaman, R. Role of the electron spin polarization in water splitting. *J. Phys. Chem. Lett.* **2015**, *6*, 4916−4922.
(14) Makarov, V. I.; Khmelinskii, I. V. Magnetic Field Influence on Dynamics of Singlet-Triplet Conversion. *Advances in Chemical Physics*; John Wiley & Sons, 2001; Vol. *118*, pp 45−98.
(15) Colvin, M. T.; Ricks, A. B.; Scott, A. M.; Smeigh, A. L.; Carmieli, R.; Miura, T.; Wasielewski, M. R. Magnetic field-induced switching of the radical-pair intersystem crossing mechanism in a donor−bridge−acceptor molecule for artificial photosynthesis. *J. Am. Chem. Soc.* **2010**, *133*, 1240−1243.
(16) Kattnig, D. R.; Evans, E. W.; Déjean, V.; Dodson, C. A.; Wallace, M. I.; Mackenzie, S. R.; Timmel, C. R.; Hore, P. J. Chemical amplification of magnetic field effects relevant to avian magnetoreception. *Nat. Chem.* **2016**, *8*, 384−391.
(17) Harkins, T.; Grissom, C. Magnetic field effects on B12 ethanolamine ammonia lyase: evidence for a radical mechanism. *Science* **1994**, *263*, 958−960.
(18) Woodward, J. R. Radical pairs in solution. *Prog. React. Kinet. Mech.* **2002**, *27*, 165−207.
(19) Rubinstein, I.; Martin, C. R.; Bard, A. J. Electrogenerated chemiluminescent determination of oxalate. *Anal. Chem.* **1983**, *55*, 1580−1582.
(20) Kazemzadeh, A.; Moztarzadeh, F. Optical oxalate detector based on chemical modification of a polymer film. *Sens. Actuators, B* **2005**, *106*, 832−836.
(21) Tang, L.; Park, J.; Kim, H.-J.; Kim, Y.; Kim, S. J.; Chin, J.; Kim, K. M. Tight binding and fluorescent sensing of oxalate in water. *J. Am. Chem. Soc.* **2008**, *130*, 12606−12607.
(22) Hu, M.; Feng, G. Highly selective and sensitive fluorescent sensing of oxalate in water. *Chem. Commun.* **2012**, *48*, 6951−6953.
(23) Wilson, D.; Liedtke, R. Modified enzyme-based colorimetric assay of urinary and plasma oxalate with improved sensitivity and no ascorbate interference: reference values and sample handling procedures. *Clin. Chem.* **1991**, *37*, 1229−1235.
(24) Sancenón, F.; Martínez-Máñez, R.; Miranda, M. A.; Seguí, M.-J.; Soto, J. Towards the development of colorimetric probes to discriminate between isomeric dicarboxylates. *Angew. Chem.* **2003**, *115*, 671−674.
(25) Gelot, M. A.; Lavoue, G.; Belleville, F.; Nabet, P. Determination of oxalates in plasma and urine using gas chromatography. *Clin. Chim. Acta* **1980**, *106*, 279−285.
(26) Geng, X.; Zhang, S.; Wang, Q.; Zhao, Z. Determination of organic acids in the presence of inorganic anions by ion chromatography with suppressed conductivity detection. *J. Chromatogr. A* **2008**, *1192*, 187−190.
(27) Robertson, W. G.; Scurr, D. S.; Smith, A.; Orwell, R. L. The determination of oxalate in urine and urinary calculi by a new ion-chromatographic technique. *Clin. Chim. Acta* **1982**, *126*, 91−99.
(28) Yriberri, J.; Posen, S. A semi-automatic enzymic method for estimating urinary oxalate. *Clin. Chem.* **1980**, *26*, 881−884.
(29) Hönow, R.; Hesse, A. Comparison of extraction methods for the determination of soluble and total oxalate in foods by HPLC-enzyme-reactor. *Food Chem.* **2002**, *78*, 511−521.
(30) Pan, H.; Shen, Y.; Duan, J.; Lu, K.; Hu, B. Spin-dependent deprotonation induced giant magnetocurrent in electrochemical cells. *Phys. Chem. Chem. Phys.* **2016**, *18*, 9897−9901.
(31) Hinds, G.; Coey, J. M. D.; Lyons, M. E. G. Influence of magnetic forces on electrochemical mass transport. *Electrochem. Commun.* **2001**, *3*, 215−218.
(32) Chang, M.-M.; Saji, T.; Bard, A. J. Electrogenerated chemiluminescence. 30. Electrochemical oxidation of oxalate ion in the presence of luminescers in acetonitrile solutions. *J. Am. Chem. Soc.* **1977**, *99*, 5399−5403.
(33) Bos, R.; Tonkin, S. A.; Hanson, G. R.; Hindson, C. M.; Lim, K. F.; Barnett, N. W. In Search of a Chemiluminescence 1,4-Dioxy Biradical. *J. Am. Chem. Soc.* **2009**, *131*, 2770−2771.
(34) Kanoufi, F.; Bard, A. J. Electrogenerated Chemiluminescence. 65. An Investigation of the Oxidation of Oxalate by Tris(polypyridine) Ruthenium Complexes and the Effect of the Electrochemical Steps on the Emission Intensity. *J. Phys. Chem. B* **1999**, *103*, 10469−10480.
(35) Lu, M.-C.; Whang, C.-W. The role of direct oxalate oxidation in electrogenerated chemiluminescence of poly(4-vinylpyridine)-bound Ru(bpy)2Cl+/oxalate system on indium tin oxide electrodes. *Anal. Chim. Acta* **2004**, *522*, 25−33.

DOI: 10.1021/acs.jpcc.8b04193
J. Phys. Chem. C 2018, 122, 19880−19885

(36) Rubinstein, I.; Bard, A. J. Electrogenerated chemiluminescence. 37. Aqueous ecl systems based on tris(2,2'-bipyridine)ruthenium(2+) and oxalate or organic acids. *J. Am. Chem. Soc.* **1981**, *103*, 512−516.
(37) Flyunt, R.; Schuchmann, M. N.; von Sonntag, C. A Common Carbanion Intermediate in the Recombination and Proton-Catalysed Disproportionation of the Carboxyl Radical Anion, CO2.−, in Aqueous Solution. *Chem.—Eur. J.* **2001**, *7*, 796−799.
(38) Neta, P.; Simic, M.; Hayon, E. Pulse radiolysis of aliphatic acids in aqueous solutions. I. Simple monocarboxylic acids. *J. Phys. Chem.* **1969**, *73*, 4207−4213.
(39) Lin, M.; Katsumura, Y.; Muroya, Y.; He, H.; Miyazaki, T.; Hiroishi, D. Pulse radiolysis of sodium formate aqueous solution up to 400°C: Absorption spectra, kinetics and yield of carboxyl radical CO2−. *Radiat. Phys. Chem.* **2008**, *77*, 1208−1212.
(40) Janik, I.; Tripathi, G. N. R. The nature of the CO2− radical anion in water. *J. Chem. Phys.* **2016**, *144*, 154307.
(41) Silvi, B.; Savin, A. Classification of chemical bonds based on topological analysis of electron localization functions. *Nature* **1994**, *371*, 683−686.
(42) Turro, N. J. Influence of nuclear spin on chemical reactions: Magnetic isotope and magnetic field effects (A Review). *Proc. Natl. Acad. Sci. U.S.A.* **1983**, *80*, 609−621.
(43) McLauchlan, K. A.; Steiner, U. E. Invited article. *Mol. Phys.* **1991**, *73*, 241−263.
(44) Hu, B.; Yan, L.; Shao, M. Magnetic-field effects in organic semiconducting materials and devices. *Adv. Mater.* **2009**, *21*, 1500−1516.
(45) Rodgers, C. T. Magnetic field effects in chemical systems. *Pure Appl. Chem.* **2009**, *81*, 19−43.
(46) Brocklehurst, B. Magnetic fields and radical reactions: recent developments and their role in nature. *Chem. Soc. Rev.* **2002**, *31*, 301−311.

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